

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the following homogeneous second-order differential equation:

$$4y'' + 8y' + 3y = 0$$

(a) Find the general solution of the given differential equation in the form of $y(x) = c_1 y_1(x) + c_2 y_2(x)$. Also verify the Wronskian $W(y_1, y_2) \neq 0$ for any x .

Solution: Characteristic equation:

$$4r^2 + 8r + 3 = 0$$

$$(2r+3)(2r+1) = 0$$

$$r_1 = -\frac{3}{2}, r_2 = -\frac{1}{2}$$

$$\Rightarrow y_1(x) = e^{-\frac{3}{2}x}, y_2(x) = e^{-\frac{1}{2}x}$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-\frac{3}{2}x} & e^{-\frac{1}{2}x} \\ -\frac{3}{2}e^{-\frac{3}{2}x} & -\frac{1}{2}e^{-\frac{1}{2}x} \end{vmatrix} = -\frac{1}{2}e^{-2x} - \left(-\frac{3}{2}e^{-2x}\right) = e^{-2x} \neq 0$$

Thus the general solution is

$$y(x) = c_1 e^{-\frac{3}{2}x} + c_2 e^{-\frac{1}{2}x}$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 2, y'(0) = -5.$$

Solution: $y(x) = c_1 e^{-\frac{3}{2}x} + c_2 e^{-\frac{1}{2}x}$

$$y'(x) = -\frac{3}{2}c_1 e^{-\frac{3}{2}x} - \frac{1}{2}c_2 e^{-\frac{1}{2}x}$$

$$\begin{cases} y(0) = 2 \Rightarrow c_1 + c_2 = 2 \\ y'(0) = -5 \Rightarrow -\frac{3}{2}c_1 - \frac{1}{2}c_2 = -5 \end{cases} \Rightarrow \begin{cases} c_1 = 4 \\ c_2 = -2 \end{cases}$$

$$\Rightarrow y(x) = 4e^{-\frac{3}{2}x} - 2e^{-\frac{1}{2}x}$$